

Proteoform Analyzer

User Manual

An AI-powered tool to analyze the effects of mutations, PTMs, and proteoforms on proteins in different oligomeric states

Proteoform Analyzer v1.0

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1 Introduction

Proteoform Analyzer quantifies how mutations, post-translational modifications (PTMs), and their combinations (proteoforms) affect a protein complex: structure, stability, dynamics, pockets, and druggability. It runs as a local web GUI (Gradio) or as a command-line pipeline, and every step executes on your own machine — there is no remote/HPC dependency.

The pipeline consists of 14 steps, each selectable individually:

Step	What it does
sequence	Sequence + mutation retrieval from UniProt.
structure	Structure preparation: Boltz-2 folding from sequence (hosted API → local binary → graft fallback).
ptm	PTM generation with PTM-Psi/ptmpsi (guided builder or UniProt auto-fetch).
tmscore	TM-score structural comparison with TM-align.
pcn	Protein Contact Network analysis (PCN-Miner): centrality, communities, WT-vs-variant deltas.
md	Molecular dynamics with OpenMM or GROMACS (RMSD/RMSF).
esm2	ESM2 embeddings + UMAP variant classification.
docking	Docking with AutoDock Vina, Boltz-2 co-folding, or DiffDock.
ligand_design	Ligand/binder design with DiffSBDD or BoltzGen.
ddg	$\Delta\Delta G$ stability prediction: ThermoMPNN / ESM2 zero-shot for mutations, FoldX mimetic scoring for PTMs (optional; requires a user-installed FoldX binary, Section 2.2).
proteoform	Proteoform generation (pairwise mutation+PTM combinations).
pocket	Binding-site prediction (P2Rank/reference/alpha-sphere) + pocket drift.
impact_score	Composite proteoform impact score.
antibody	De novo antibody/nanobody design with RFAntibody (opt-in).

2 Installation

2.1 Local installation

Requirements: Python 3.11, Java (for P2Rank), and — for the antibody step — an NVIDIA GPU with recent drivers plus uv (<https://docs.astral.sh/uv/>).

```
unzip proteoform_analyzer.zip
cd proteoform_analyzer_zip/proteoform_analyzer
python3.11 -m venv .venv && source .venv/bin/activate
pip install -r pre-requirements.txt
pip install -r requirements.txt
python app.py                # GUI on http://0.0.0.0:7860
```

The first time the antibody step runs, it bootstraps the vendored RFAntibody environment automatically (uv sync inside proteoform_analyzer/_vendored/RFantibody) and downloads the three model weights (≈ 750 MB) if they are missing. Both actions are one-time; subsequent runs reuse the environment and weights.

2.2 FoldX installation (optional, required for PTM $\Delta\Delta G$)

FoldX (<https://foldxsuite.cemmm.at>) is used as an *optional* third tier of the ddg step to score the stability impact of **PTMs** via accepted mimetic substitutions (phosphorylation SER/THR→GLU,

acetylation LYS→GLN). Mutation $\Delta\Delta G$ does *not* require FoldX (ThermoMPNN / ESM2 zero-shot cover it), and every other step runs without it.

FoldX is **not bundled** with Proteoform Analyzer and cannot be redistributed: it is free only for *academic* use under a custom licence, so each user must obtain it personally. Installation takes about five minutes:

1. **Request access.** Go to <https://foldxsuite.cemmm.at>, click *Download*, and complete the academic licence agreement (name, institution, e-mail). The download link is sent by e-mail or revealed after approval, depending on the current policy of the FoldX team.
2. **Download the suite.** From the download page pick *FoldX (Linux 64-bit)* — e.g. `FoldX5_Linux64.zip` — and, if offered separately, the `rotabase.txt` parameter file (recent builds bundle it inside the archive).
3. **Unpack and make the binary executable.**

```
unzip FoldX5_Linux64.zip -d ~/foldx
chmod +x ~/foldx/foldx
# rotabase.txt must sit next to the binary (or use --foldx-rotabase):
ls ~/foldx          # foldx  rotabase.txt
```

4. **Point Proteoform Analyzer at it** — any *one* of:

- *Environment variable* (recommended; picked up by GUI and CLI):

```
export FOLDX_BINARY="$HOME/foldx/foldx"
```

- *GUI*: 1. **Setup** → **Engine choices** → **FoldX binary path** (Figure 5).
 - *CLI*: `--foldx-binary /foldx/foldx` (optionally `--foldx-rotabase` and `--foldx-n-runs N` to average BuildModel replicates).
5. **Verify.** Run any analysis with PTMs enabled; the `ddg` step log then reports FoldX rows (method = `foldx_mimetic`) in `results/<run>/ddg/ddg_summary.csv`. If no binary resolves, the tier is skipped with an explicit log message and everything else is unaffected.

How the PTM scoring works. FoldX parameterises only the 20 standard amino acids and has no energy terms for modified residues, so each PTM is scored as its accepted *mimetic substitution*: phosphorylation as SER/THR→GLU (phosphomimetic) and acetylation as LYS→GLN (acetyl-mimetic). Each site is mapped onto the WT structure with residue-name verification (mature numbering first, then UniProt), scored on the WT background, and every mutation × PTM proteoform is scored with the mutation and the mimetic combined in a single BuildModel line after RepairPDB pre-processing. PTMs without an accepted mimetic are skipped with an explicit warning — never silently approximated.

2.3 Docker installation (recommended)

The repository ships an all-in-one GPU-capable image. Prerequisites: Docker, and — for GPU steps — the NVIDIA Container Toolkit.

```
# Build (weights baked in, ~15 GB image)
docker build -t proteoform-analyzer .

# Build without RFAntibody weights (smaller; downloaded on first use)
docker build --build-arg DOWNLOAD_WEIGHTS=false -t proteoform-analyzer .

# Run (CPU)
```

```
docker run -p 7860:7860 proteoform-analyzer

# Run (GPU)
docker run --gpus all -p 7860:7860 proteoform-analyzer

# Persist results on the host
docker run --gpus all -p 7860:7860 \
  -v $(pwd)/results:/app/results proteoform-analyzer
```

Then open <http://localhost:7860>. To use the hosted Boltz API inside the container, pass `-e BOLTZ_API_KEY=...`. To run the CLI instead of the GUI, override the command, e.g. `docker run ... proteoform-analyzer python3.11 -m proteoform_analyzer.cli --help`.

FoldX in Docker. For the same licensing reason FoldX is not baked into the image either. Install it on the *host* and mount it into the container at run time:

```
docker run --gpus all -p 7860:7860 \
  -v $(pwd)/results:/app/results \
  -v $HOME/foldx:/opt/foldx:ro \
  -e FOLDX_BINARY=/opt/foldx/foldx \
  proteoform-analyzer
```

3 Graphical interface reference

Start the GUI with `python app.py` and open <http://localhost:7860>. The interface has three tabs: **1. Setup** (configure the analysis), **2. Run** (execute and monitor), and **3. Results** (browse outputs).

3.1 Tab 1: Setup

3.1.1 Preset and protein definition

Preset configuration	One of nine curated presets — <i>Hemoglobin</i> / <i>TTR</i> / <i>p53</i> , each in <i>Fast</i> , <i>Standard</i> , and <i>Production</i> flavors. Selecting a preset fills every field below; all fields remain editable afterwards. <i>Fast</i> presets cap mutations/PTMs at 10 for a quick smoke test; <i>Production</i> presets remove the caps (Figure 1).
UniProt ID(s)	Comma-separated UniProt accessions, one per distinct subunit sequence (e.g. P69905,P68871 for hemoglobin α/β).
Number of subunits	Total subunit count of the oligomer (1 = monomer, 4 = tetramer, ...).
Stoichiometry	Comma-separated copy counts matching the UniProt ID order (e.g. 2,2 = two α + two β).
Proteoform mode	single : each variant carries one modification; pairwise : mutations and PTMs are combined pairwise per subunit.
Max proteoforms	Cap on the number of generated proteoforms (10–500).
Mutations (pipe-separated per subunit)	Mutations in the form E7V D95H D75A — space-separated within a subunit, -separated across subunits, in UniProt numbering.

Max mutations	Maximum number of mutations (and, in auto-fetch mode, PTMs) kept per analysis (1–50).
Structure source	boltz2 : Boltz-2 folds structures from sequence (hosted API → local binary → graft fallback).
Reference PDB ID	Optional experimental structure (e.g. 1A3N) used <i>only</i> to align known binding sites; it is not a structure source.

Figure 1: Setup tab after loading the Hemoglobin — Fast preset.

3.1.2 PTM settings (guided builder)

The PTM accordion replaces free-text PTM entry with a builder that only offers existing, chemically possible PTMs:

1. **Load modifiable residues from UniProt** — fetches the UniProt sequences for the IDs above and lists only residues that can carry a modelable PTM (including position 1, which can carry N-terminal alpha-acetylation). A status line reports how many modifiable residues were found per UniProt ID (Figure 2).
2. **Residue** — dropdown of modifiable residues, shown as UNIPROT_ID · RESIDUE+POSITION (e.g. P69905 · MET1; Figure 3).
3. **PTM type** — dropdown filtered to the PTMs chemically compatible with the selected residue (e.g. a serine offers phosphorylation; a cysteine offers the nine cysteine-specific PTMs; position 1 offers alpha-acetylation). PTMs *observed in UniProt* at that exact residue are listed first and tagged.
4. **Add PTM / Clear all** — append the selected pair to the table, or empty the table (Figure 4).

5. **Selected PTM sites** — editable table of `residue` | `ptm_type` | `uniprot_id` rows; rows can be deleted via the row menu.

Leave the table empty to let the pipeline auto-fetch the PTMs observed in UniProt for each subunit, capped at the *Max mutations* value. If a protein has no known modelable PTMs, the PTM step is skipped automatically with a clear log message.

The screenshot displays the PTM builder interface. At the top, there are input fields for 'Mutations (pipe-separated per subunit)' and 'Max mutations' (a slider from 0 to 50). Below these is a 'Fetch from UniProt' button. The 'Structure source' section shows 'boltz2' selected. The 'Reference PDB ID (binding-site alignment only)' field contains '1A3N'. A section titled 'Mature chain (proteolytic processing)' is collapsed. The 'PTM settings' section is expanded, showing instructions and a 'Load modifiable residues from UniProt' button. Below this, a status line reports 'P69905: 61 modifiable residues — P68871: 56 modifiable residues'. The 'Residue' dropdown is set to 'P69905 - MET1' and the 'PTM type' dropdown is set to 'alpha-acetylation'. An 'Add PTM' button is visible. Below the 'Selected PTM sites' section, a table with columns 'residue', 'ptm_type', and 'uniprot_id' is shown, currently empty. The 'Engine choices' section at the bottom shows 'MD engine' selected and 'MD production steps' set to 5000.

Figure 2: PTM builder after loading modifiable residues: the status line reports 61 modifiable residues for P69905 and 56 for P68871; the dropdowns are populated.

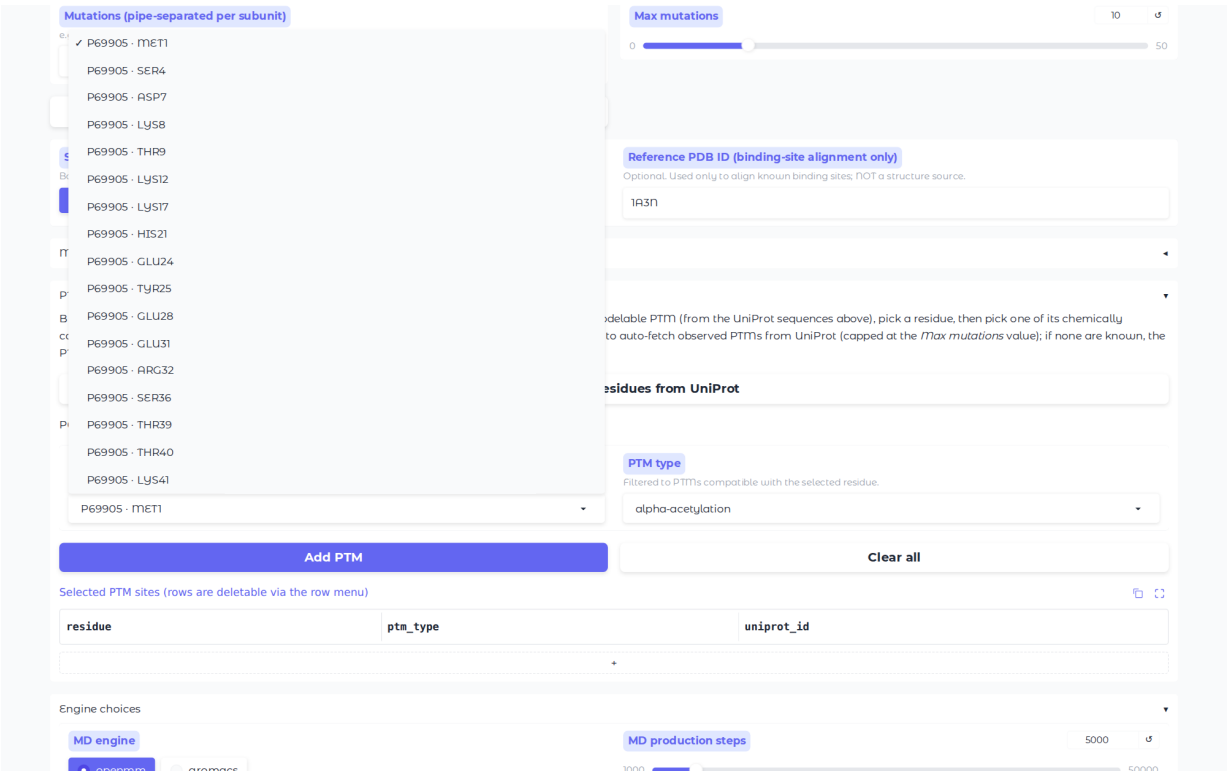


Figure 3: Residue dropdown: only PTM-competent residues are listed (P69905 · MET1 selected).

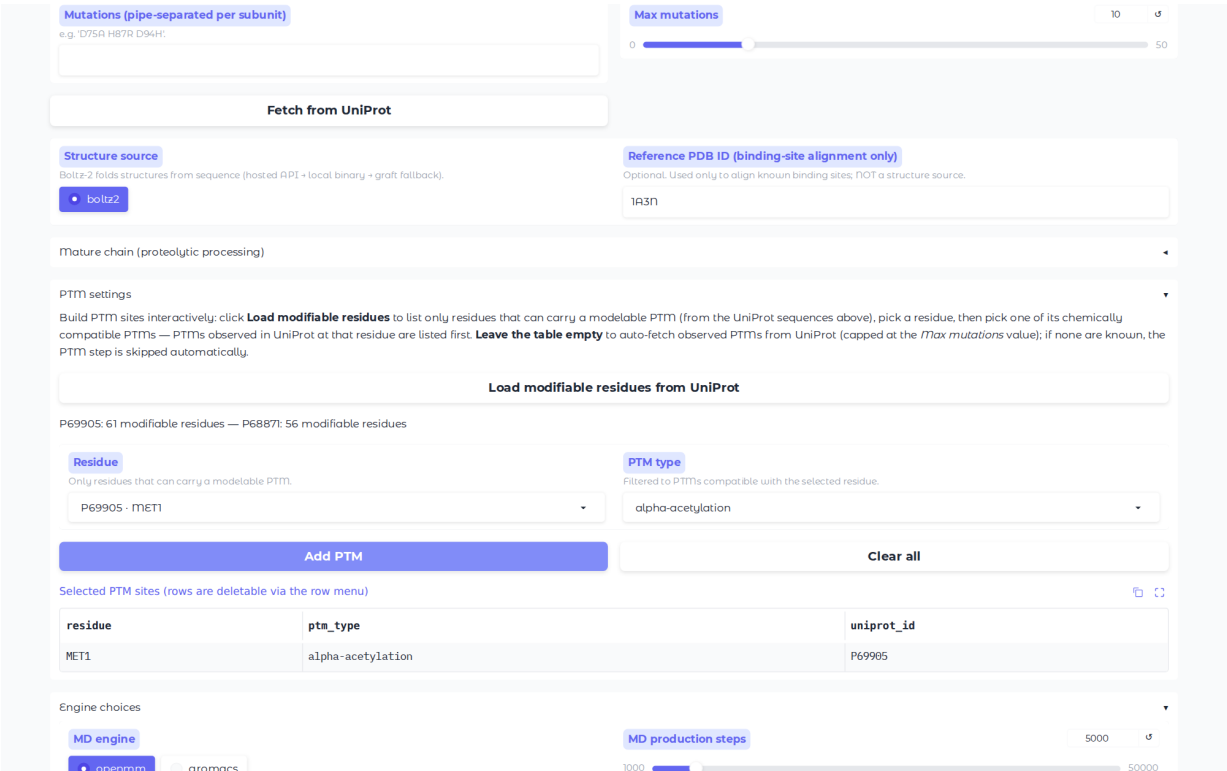


Figure 4: One PTM site added to the table: MET1 | alpha-acetylation | P69905.

3.1.3 Engine choices

MD engine openmm (default) or gromacs.

MD production steps	Production-run length (1 000–50 000 steps; default 5 000).
Docking engine	vina (AutoDock Vina, kcal/mol), boltz2 (co-folds receptor+ligand; confidence score, not kcal/mol), or diffdock .
Ligand / binder design engine	diffsbdd (small-molecule design) or boltzgen (protein binder design; Boltz API or local install).
Binding-site prediction method	auto (default), reference (fixed box from a known ligand site; no pocket volume/drift), p2rank , alphasphere , or user .
Ensemble docking	Dock against MD snapshots instead of a single static structure.
Boltz API key	Optional key for the hosted Boltz backend. Backend precedence is always: hosted API key → local install → graft fallback (folding only).
FoldX binary path	Path to a local FoldX executable (FoldX 4/5, academic licence — see Section 2.2). Overrides \$FOLDX_BINARY; rotabase.txt is taken from the binary’s directory. When set, PTMs are additionally scored for stability ($\Delta\Delta G$) as mimetic substitutions; when blank, PTM stability scoring is skipped (Figure 5).

The screenshot displays the configuration interface for Proteoform Analyzer. It includes several sections with settings:

- Docking engine:** Radio buttons for **vina** (selected), **boltz2**, and **diffdock**.
- Ligand / binder design engine:** Radio buttons for **diffsbdd** (selected) and **boltzgen**.
- Binding-site prediction method:** Radio buttons for **auto** (selected), **reference**, **p2rank**, **alphasphere**, and **user**.
- Ensemble docking (MD snapshots):** An unchecked checkbox.
- Boltz-2 backend:** A text box for the Boltz API key.
- Boltz API key:** A text box for the Boltz API key.
- Prefer local Boltz:** An unchecked checkbox.
- Allow graft fallback:** A checked checkbox.
- FoldX (PTM $\Delta\Delta G$):** A text box for the FoldX binary path.
- Antibody design (RFAntibody; opt-in):** A section with a sub-section for **Pipeline steps to run** containing checkboxes for **sequence**, **structure**, **ptm**, **tm_score**, **pcn**, **md**, **esm2**, **docking**, **ligand_design**, **ddg**, **proteoform**, **pocket**, **impact_score**, and **antibody**.

Figure 5: Engine choices: docking/design engines, Boltz backend, and the new **FoldX binary path** field (bottom).

3.1.4 (Optional) Antibody design

Open **Antibody design (RFAntibody; opt-in)** (Figure 6) to enable de novo nanobody design against the predicted B-cell epitopes of the target. This step needs a GPU; on a CPU-only

machine it skips cleanly. The hotspot source defaults to `bcell` — switch to `user` to target specific residues.

Docking engine
boltz2: co-folds receptor+ligand (confidence score, not kcal/mol).
☒ vina ☐ boltz2 ☐ diffdock

Ligand / binder design engine
diffsbdd: small-molecule design; boltzgen: protein binder design (Boltz API + local install + none).
☒ diffsbdd ☐ boltzgen

Binding-site prediction method
reference: uses a fixed box from a known ligand site (no pocket volume/drift — see warning in Results).
☒ auto ☐ reference ☐ p2rank ☐ alphasphere ☐ user

Boltz-2 backend — structure folding, co-folding docking (boltz2), and binder design (boltzgen) all use one backend, chosen in this order: **hosted API key** + **local install** + **graft fallback** (folding only). Leave everything blank to use the offline graft fallback. Local binder design runs BoltzGen if installed.

Boltz API key
Official opiboltz.bio key. Overrides \$BOLTZ_API_KEY. Enables folding, docking, and binder design via the hosted API (no GPU needed).

☐ Prefer local Boltz
Use a local 'boltz/boltzgen' install before the API.

☒ Allow graft fallback
If no folding backend exists, build backbone-identical structures by PTM-Psi grafting (TM-score = 1.0; no structural/pocket/docking signal). Uncheck to hard-skip.

FoldX (PTM $\Delta\Delta G$) — optional, academic-licensed. When a binary is provided, PTMs are scored for stability ($\Delta\Delta G$) as mimetic substitutions (phospho SER/THR+GLU, acetyl LYS+GLN) on the WT background and on each mutation + PTM proteoform. Leave blank to skip PTM stability scoring.

FoldX binary path
Path to the FoldX executable. Overrides \$FOLDX_BINARY. rotabase.txt is taken from the binary's directory (or --foldx-rotabase).

Antibody design (RFAntibody; opt-in)
De novo antibody/nanobody design against a binding site (RFDiffusion -> ProteinMPNN -> RF2). Enabling this adds the **antibody** step automatically. Requires a **local RFAntibody install** (GPU strongly recommended); set the checkout path via `config.antibody.local_rfantibody_dir` (or the `--antibody-rfantibody-dir` CLI flag). Without it, the step skips cleanly with instructions.

☒ Enable antibody design

Framework
☒ nanobody ☐ scfv

Number of designs
20

Hotspot source
Target hotspot residues (comma-sep)

Figure 6: Antibody design accordion (nanobody framework, 20 designs, hotspot source `bcell`).

3.2 Tab 2: Run

Switch to **2. Run** and click **Run pipeline** (Figure 7). Follow progress in the live log and the per-step status table; skipped steps (if any) carry an explanatory message.

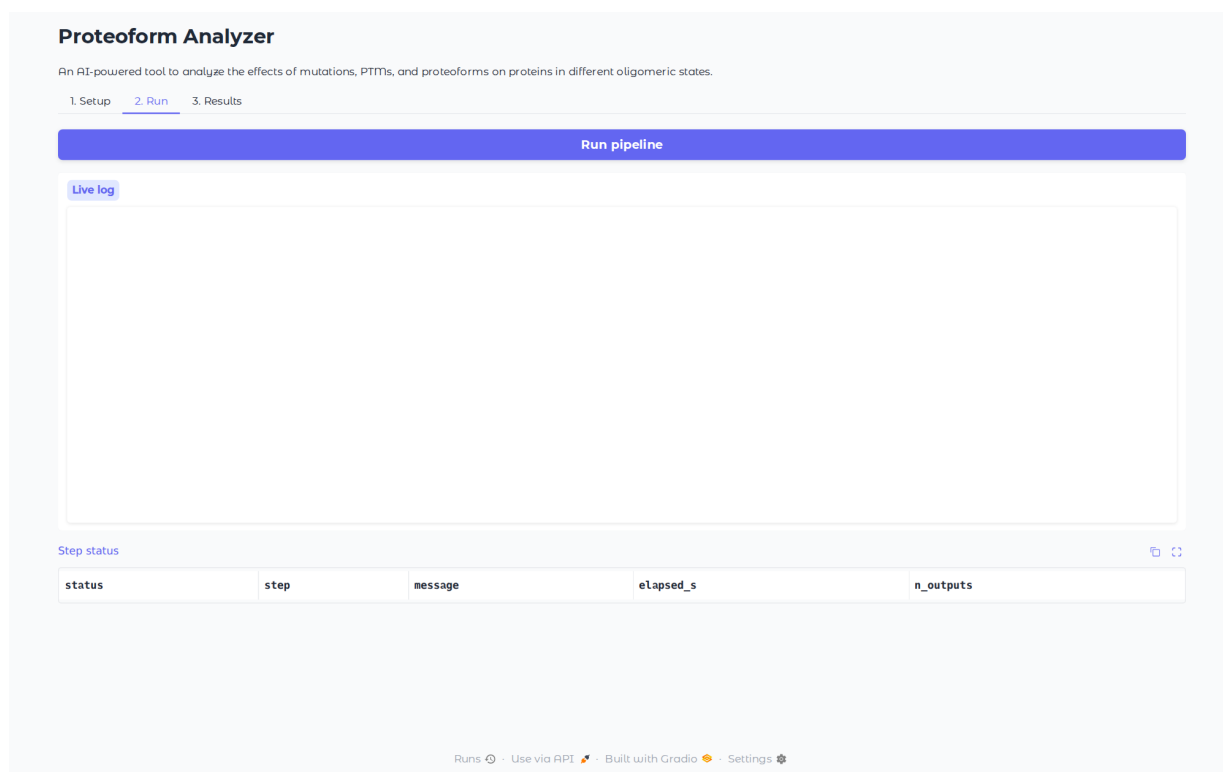


Figure 7: Run tab: **Run pipeline**, live log, and step status.

3.3 Tab 3: Results

Switch to **3. Results** and click **Refresh all results** (Figure 8). Open each accordion for impact scores, pockets, TM-scores, docking, MD overlays, contact networks, ESM2+UMAP, $\Delta\Delta G$, designed ligands, predicted antibodies, and the 3D viewer. All files are also written to `results/<run_name>/` on disk.

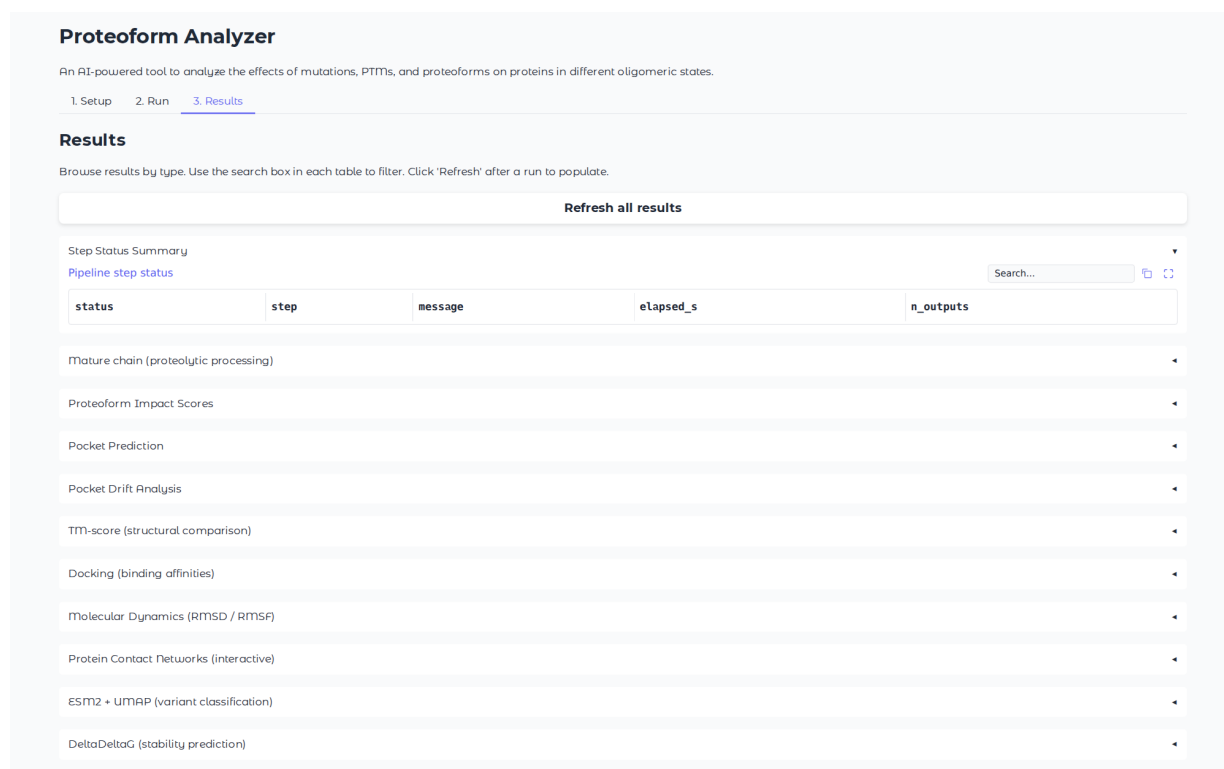


Figure 8: Results tab with one accordion per analysis type.

4 Example results

This section shows what the pipeline produces, using the three bundled example runs shipped in the repository under `results/`: TTR missense variants (`ttr_mutations`), TTR PTMs (`ttr_ptms`), and the hemoglobin heterotetramer (`hemoglobin`). All numbers come from the CSV files written by the pipeline; they illustrate typical output, not biologically validated conclusions.

4.1 TTR missense variants (87 mutations, homotetramer)

The TTR — Fast preset folds the mature transthyretin tetramer (UniProt P02766, residues 21–147) with Boltz-2, enumerates 87 clinically relevant missense variants, and runs the full workflow (TM-align, PCN, OpenMM MD, ESM2+UMAP, Vina docking of tafamidis, ThermoMPNN $\Delta\Delta G$, impact scores, RFAntibody).

- **Stability ($\Delta\Delta G$, ThermoMPNN):** 87 predictions ranging from -0.36 (M33I, mildly stabilising) to $+3.29$ kcal/mol (G73E, strongly destabilising); glycine substitutions (G67E $+2.93$, L75Q $+2.90$) dominate the top of the ranking — consistent with the amyloidogenic sensitivity of the TTR β -sheet region.
- **Docking (tafamidis, Vina):** 88 structures docked in a -5.63 to -4.60 kcal/mol window; F53I/F53L bind most favourably (-5.63).
- **Impact score:** the top-ranked proteoform is `Mut_P02766_A65D-tetramer` (composite 0.60), driven by a large binding shift (0.87) plus dynamics (0.70) and network (0.67) components (Table 2).

Table 2: Top-5 Proteoform Impact Scores, TTR missense-variant run (`results/ttr_mutations/impact_scores/`). Components are normalised to $[0, 1]$; the sequence component was dropped as degenerate in this run and its weight renormalised over the remaining components.

Proteoform	Structural	Binding	Dynamics	Network	Composite
Mut_A65D-tetramer	0.043	0.875	0.697	0.667	0.601
Mut_F53L-tetramer	0.035	1.000	0.662	0.333	0.576
Mut_F53I-tetramer	0.040	1.000	0.684	0.000	0.523
Mut_G67R-tetramer	0.021	0.741	0.687	0.333	0.487
Mut_E81G-tetramer	0.044	0.680	0.727	0.333	0.480

4.2 TTR PTMs (7 modification sites)

The `ttr_ptms` run builds seven PTM proteoforms on the WT tetramer (phosphorylation at THR23, SER72, SER137, THR138, THR25, TYR125 and nitrosylation at CYS10), then runs TM-align, PCN, MD, docking and impact scoring on each. Phosphorylation at **THR23** ranks first (composite 0.59), combining the largest binding shift (tafamidis affinity changes by 0.28 of the observed maximum) with the largest community-count change; the seven tafamidis docking scores span a narrow -5.05 to -5.01 kcal/mol band. With a FoldX binary configured (Section 2.2), each site would additionally receive a $\Delta\Delta G$ row (`method = foldx_mimetic`) in `ddg/ddg_summary.csv`.

4.3 Hemoglobin $\alpha_2\beta_2$ heterotetramer

The Hemoglobin — Fast preset (P69905,P68871, stoichiometry 2,2) demonstrates heteromeric handling: mature-chain detection (initiator-Met removal), 54 mutation $\Delta\Delta G$ predictions (ThermoMPNN; most destabilising W15R +2.49 kcal/mol), voxelotor docking against 51 structures transferred from the co-crystal pocket (-7.65 to -7.00 kcal/mol; best D7A), and a top-ranked proteoform `Mut_P68871_W16G-tetramer` (composite 0.60). The run also produces the full plot set (`results/hemoglobin/_plots/`): impact summary, docking affinities, MD RMSD/RMSF overlays, pocket volume/drift, TM-scores, antibody rankings, and the ESM2+UMAP projection separating the α and β subunits.

5 Command-line interface

The same pipeline runs headless via `python -m proteoform_analyzer.cli`. Key options (see `--help` for the full list):

Option	Meaning
<code>--preset NAME</code>	Load one of the nine presets.
<code>--uniprot IDS</code>	Comma-separated UniProt IDs.
<code>--mutations "E7V D75A"</code>	Mutations per subunit, pipe-separated.
<code>--n-subunits N</code>	/ Oligomer definition.
<code>--stoichiometry 2,2</code>	
<code>--chain-map</code>	UniProt $\rightarrow$ chain mapping.
<code>"P69905:A,C;P68871:B,D"</code>	
<code>--max-mutations N</code>	Cap on mutations (and auto-fetched PTMs).
<code>--proteoform-mode</code>	Proteoform combination mode.
<code>single pairwise</code>	

Option	Meaning
<code>--ptm-pair RES:PTM[:UID]</code>	Explicit PTM site, repeatable, e.g. <code>--ptm-pair "SER15:phosphorylation:P04637"</code> . When omitted, observed PTMs are auto-fetched from UniProt (capped); if none exist the PTM step is skipped.
<code>--no-ptm / --no-md</code>	Disable the PTM / MD steps.
<code>--steps a,b,c</code>	Run only the listed steps.
<code>--md-engine openmm gromacs /</code>	MD configuration.
<code>--md-steps N</code>	
<code>--docking-engine vinal boltz2 diffdock</code>	Docking backend.
<code>--ligand-design-engine diffsbdd boltzgen</code>	Design backend.
<code>--binding-site-method M</code>	<code>auto reference p2rank alphasphere user</code> .
<code>--ensemble-docking</code>	Dock against MD snapshots.
<code>--foldx-binary PATH</code>	New. Local FoldX executable for PTM $\Delta\Delta G$ (Section 2.2); falls back to <code>\$FOLDX.BINARY</code> or <code>foldx</code> on <code>PATH</code> . If unset, PTM $\Delta\Delta G$ is skipped.
<code>--foldx-rotabase PATH</code>	Optional <code>rotabase.txt</code> location (default: next to the binary).
<code>--foldx-n-runs N</code>	FoldX <code>BuildModel numberOfRuns</code> (replicates averaged; default 1).
<code>--antibody</code>	Enable the RFAntibody step.
<code>--antibody-framework nanobody scfv</code>	Scaffold framework.
<code>--antibody-num-designs N</code>	RFdiffusion backbones (default 20).
<code>--antibody-hotspot-source S</code>	<code>bcell (default)</code> , <code>user</code> , <code>mhc_i</code> , <code>mhc_ii</code> .
<code>--antibody-hotspots "305,456"</code>	Hotspots for source <code>user</code> .
<code>--antibody-rfantibody-dir D</code>	Override vendored RFAntibody / weights locations.
<code>/ --antibody-weights-dir D</code>	
<code>--work-dir DIR</code>	Output root (default <code>results</code>).

Example — hemoglobin with two explicit PTMs, FoldX PTM $\Delta\Delta G$, and the antibody step:

```
python -m proteoform_analyzer.cli \
  --preset "Hemoglobin - Fast" \
  --ptm-pair "MET1:alpha-acetylation:P69905" \
  --ptm-pair "SER44:phosphorylation:P68871" \
  --foldx-binary ~/foldx/foldx \
  --antibody --antibody-framework nanobody \
  --work-dir results
```

6 Troubleshooting

- PTM step skipped** The protein has no UniProt-observed modelable PTMs and the builder table is empty — expected behavior. Add sites manually with the builder or `--ptm-pair`.
- PTM type list is short** By design: only PTMs chemically compatible with the selected residue are offered. Cysteine has the longest list (nine PTMs); most residues have one or two.

FoldX tier skipped / no foldx rows

No binary resolved. Check, in order: the **FoldX binary path** field, `$FOLDX_BINARY`, and `foldx` on `PATH`; that the file exists and is executable (`chmod +x`); and that `rotabase.txt` sits next to the binary (or pass `--foldx-rotabase`). The `ddg` log line reports which resolution failed. Remember: only PTM $\Delta\Delta G$ needs FoldX; mutation $\Delta\Delta G$ uses ThermoMPNN/ESM2.

FoldX PTM skipped with warning

The PTM has no accepted mimetic substitution (only phosphorylation and acetylation are mimetic-scored today). This is explicit by design — the pipeline never silently approximates an unsupported modification.

Antibody step fails at stage 1

Open `<results>/antibody/logs/stage1_rfdiffusion.log`. Usually the GPU is unavailable or the first-time bootstrap (`uv sync`, weight download) was interrupted — rerun; the bootstrap resumes. If the log shows “*Non-positive determinant ... in rotation matrix*”, the target or framework PDB contained alternate conformations (altlocs) or missing backbone atoms; the step collapses altlocs automatically when building `target.pdb`, so rerun — or point the step at a cleaned/AlphaFold-DB structure.

Broken vendored venv If the shipped RFAntibody `.venv` was created on another machine, the step detects it, removes it, and re-bootstraps automatically.

Boltz steps skip No API key and no local install: folding falls back to graft; co-folding docking and BoltzGen skip cleanly. Set `BOLTZ_API_KEY` or install Boltz locally.

Docker GPU not visible

Install the NVIDIA Container Toolkit and run with `--gpus all`.

Where are the files? Everything lands in `<work-dir>/<run-name>/`; antibody stage logs in `antibody/logs/`, designed PDBs in `antibody/designs/`, scores in `antibody/scores.tsv`, impact scores in `impact_scores/`, $\Delta\Delta G$ in `ddg/`.